

Modelling irruptions and population dynamics of the great spotted woodpecker – joint effects of density and cone crops

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Irruptive migrants are partially migrating species showing pronounced temporal variation in the number of migrants. Occasional irruptions are often explained by increased population density (per area) or lack of food. Similarly to population dynamics, these explanations are not mutually exclusive, but probably act in concert. Here we do a parallel analysis on the migration intensity and population dynamics of an irruptive migratory bird species, the great spotted woodpecker *Dendrocopos major*, in southern Finland. Both analyses include simultaneous effects of breeding density and cone crops of Scots pine *Pinus sylvestris* and Norway spruce *Picea abies*. A novel model for statistical analyses of irruptive migration intensity is developed. Population dynamics are investigated using log-linear state-space models. Woodpecker migration intensity is well explained by lack of spruce cones and increases proportionally to population density. Population dynamics is most successfully described with a density dependent model, where the natural logarithm of spruce cone availability previous autumn has a strong positive effect on population growth. This is likely to be due to lowered winter mortality and lowered emigration rates resulting from plentiful food availability. Although the species uses pine cones as a food resource in winter, the estimated impacts of pine cone crops on migration intensity and population dynamics are low, probably due to low annual variation in crop comparing to spruce cones. Large irruptions also tend to occur earlier in season, often before cones are a crucial food resource for the woodpeckers, suggesting that the woodpeckers use the amount of maturing cones as a cue for upcoming conditions. Our results show that the ecological role of different resources can differ considerably from the expected pattern based on the proportion of the resources in a species' diet. Advantages of examining patterns of bird migration, population density and dynamics in unison are apparent.

Emigration and immigration are fundamental processes in population dynamics and can be determined by local environmental conditions, availability of resources or population density (Bowler and Benton 2005). These factors affecting migration are typically also linked to births and deaths, the two other fundamental components of population dynamics (Royama 1992). Some animal species, especially some birds in the northern hemisphere, occasionally show massive irregular migration events, called irruptions (Reinikainen 1937, Lack 1954, Svärdsön 1957, Bock and Lepthien 1976, Alerstam 1990, Newton 2006). These events occur mainly in autumn and can be strongly synchronous among related species (Bock and Lepthien 1976). In addition to high annual variability in the intensity of migration, species with irruptive migration behaviour are typically partial migrants with poor site fidelity. Also direction, distance and timing of such migration show typically high variation compared to regular annual migration. Population dynamics of species showing irruptive movements are likely to be regulated by the interplay of population density and

resource availability, as well as the irruptive movements per se (Newton 2006). Consequently, we find it appropriate to investigate plausible factors behind irruptive movements and population dynamics in parallel.

It is well known that the triggering of irruptions is related to a lack of food, resulting from e.g. poor crops of seeds or berries, or from the crash of prey populations (Alerstam 1990, Newton 2006, Fox et al. 2009). These food resources are often distributed unevenly in space and show considerable temporal variation (Kelly 1994, Hanski et al. 2001). The intensity of irruptive migration events is negatively associated with food abundance at least in finches, tits, nutcrackers, owls and woodpeckers (Reinikainen 1937, Pynnönen 1943, Newton 1970, 2006, Eriksson 1971, Cramp 1985, Alerstam 1990, Fox et al. 2009).

In addition to food shortage, increased population density has been suggested to be the trigger of irruptive migration (Lack 1954, Alerstam 1990). Koenig and Knops (2001) suggested that population density is more important than food shortage in triggering irruptions of North American

redpolls *Carduelis flammea* and black capped chickadees *Parus atricapillus*, whereas in white-winged crossbills *Loxia leucoptera* the situation seemed to be reversed. They also concluded that irruptions of many species usually occur through a combination of high population densities, resulting from good seed crops, and subsequent poor seed crops. Newton (1970) proposed that a large population size is a prerequisite for strong irruptive movement to take place, simply by affecting the potential number of migrants. On the other hand, behavioural responses to population density might actively lead to irruptive movements. As an example, willow tits *Parus montanus* form small wintering flocks, consisting of two adults and two to six young birds. The species is mainly resident, but following high breeding success some juveniles are left outside wintering flocks and will consequently emigrate (Silverin 2003).

Food shortages and density are not mutually exclusive explanations for irruptions, but can act in concert (Lack 1954, Bock and Lephtien 1976, Koenig and Knops 2001). Firstly, population size affects the number of potential migrants. Secondly, irruptive migration events can be triggered by food, population density or a combination of both. Hence, a sound model for irruptive migration should explicitly cope with intrinsic factors (population density) and extrinsic factors (such as food supply) simultaneously. In analogy with population dynamical regulation, we refer to this as a synthetic view of irruptive migration (Turchin 1995, 1999).

The purpose of this study is to clarify the relative importance of population density and food abundance in determining the intensity of irruptive migration events and population dynamics of the great spotted woodpecker *Dendrocopos major*. The aim is to explore the mechanistic background of these phenomena, by comparing rate and ratio-dependent candidate models with different functional relationships. Despite the large number of studies linking irruption intensity to food shortage and to some extent also population density, no serious attempt to construct quantitative mechanistic models on the issue has to our knowledge been made. We propose a model for integrating the effects of food abundance and breeding density on irruptive migration. Also the relationship between timing and intensity of migration is investigated.

Material and methods

The study species and design

The great spotted woodpecker is a widely distributed bird species in Eurasia, breeding in northern Europe during May–June (Cramp 1985). It is normally resident but occasionally shows heavy irruptive migration events in boreal areas with up to tens of thousands of migrants (Eriksson 1971, Cramp 1985, Alerstam 1990). Transient woodpeckers are mainly young individuals (Hildén 1974), possibly due to their lower social status or poorer ability to find food during winter (Silverin 2003). In Fennoscandia the most important winter food resources of the great spotted woodpecker are seeds of two abundant coniferous tree species, Scots pine *Pinus sylvestris* and Norway spruce *Picea abies* (Cramp 1985).

Previously Eriksson (1971) has shown that wintering densities of great spotted woodpeckers are positively related to the relative abundance of spruce cones, but he did not believe that lack of food provided an exhaustive explanation for irruptions. Hildén (1974) suggested that irruptions are caused by crop failure of both spruce and pine, but he had no data supporting this view. Nevertheless, based on breeding surveys and trapping totals in Ottenby (Sweden), Svensson et al. (1986) suggested that irruptions are caused by a combination of population density and food shortage, but they did not measure food abundance. Also the timing of woodpecker migration shows huge annual variation (Lehikoinen and Vähätalo 2000), which could be related to food resources or migration intensity. Svårdson (1957) and Hildén (1974) suggested that heavy irruptions seem to happen earlier in season than smaller ones. Factors regulating breeding population dynamics of the species have to our knowledge not been explicitly studied using density dependent population dynamical models.

In this study, we investigate the irruptive migration intensity and population dynamics of the species in southern Finland using quantitative statistical models. The study design, involving data, analyses, response- and explanatory variables, are illustrated in Fig. 1. Annual breeding population estimates are based of national monitoring data, migration intensity and phenology estimates are based on bird observatory data, while annual food availability is based on cone monitoring schemes. These are closer described in

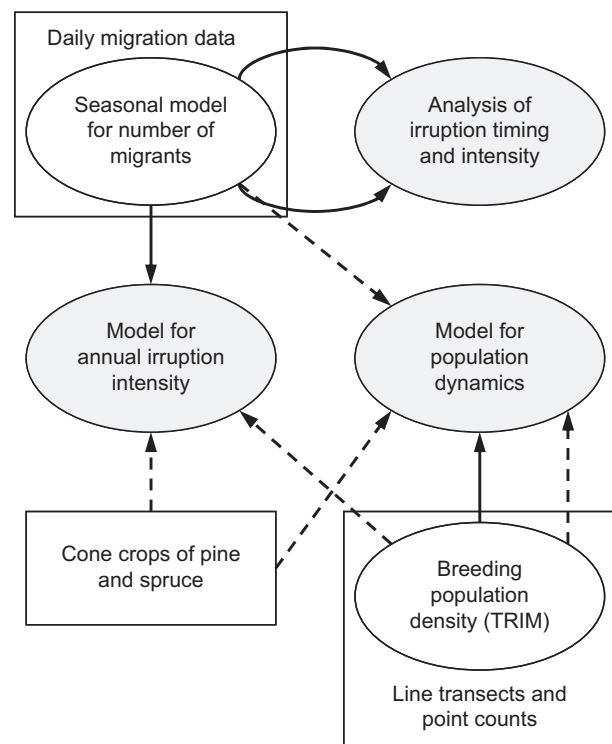


Figure 1. The study design. Open symbols indicate data, while shaded symbols represent inferential models or analyses. Squares are sources of raw data, while processes involving statistical estimation (e.g. data cleaning) are indicated with ovals. Input variables to the inferential analyses are displayed with arrows, such that solid arrows are response variables and dashed arrows are explanatory variables.

the following sections. The time series used for inferential analyses are given in Appendix 1.

Study area and breeding population data

The breeding density of woodpeckers and data on cone crops comprise a large area, consisting of the southern half of Finland (south from 64°N, area ca 400 × 500 km). Annual breeding densities of woodpeckers are based on line transect and point census counts from 1979–2007, collected as a part of the avian breeding monitoring censuses of the Finnish Museum of Natural History (Väisänen 2006). The average annual count is 132 woodpecker observations indicating 'pairs' in the Finnish census tradition (a male, a female, a pair, a group of fledglings or an inhabited nest; Koskimies and Väisänen 1991). The software TRIM (Pannekoek and van Strien 2004) is used to calculate annual density indices from transects/sites, where the same census site has been counted in at least two years. The density in the first year (1979) is scaled to one. The resulting time series is used as an estimate of the relative population density (N_i) in southern Finland and is hereafter denoted as Z_i in the descriptions of models.

Migration data

For quantifying migration intensity and timing of migration we use daily data from standardized counts on migrating and resting woodpeckers at the Hanko Bird Observatory (59°49'N, 22°54'E). Data were gathered during 1979–2007, between the dates 1 July and 10 November, a period spanning most of the woodpecker migration (Lehikoinen and Vähätalo 2000). The observatory is situated at the southwestern tip of the Hanko peninsula on the southern coast of Finland and is a well known bottleneck for bird migration in autumn, collecting birds from at least southern Finland, but probably also from e.g. northern Finland and Russia. We assume that the observed number of woodpeckers at the observatory provides a reasonably good proxy for intensity of migration originating from the study area, even though little is known about between-year variation in migration direction and other related factors affecting bird counts. The Hanko data are used to estimate both the relative annual intensity of migration and the timing of migration.

Despite the reasonably good coverage in counting, there are data missing for about 10% of the days, mainly early in season before 1995. The daily number of observed birds can vary considerably, mostly due to variation in weather conditions. The day-to-day variation can be regarded as statistical noise, when assessing large scale patterns like annual intensity or timing of migration. A robust data cleaning approach accounting for missing days and day-to-day variation is to model the expected number of migrants using seasonal distribution curves (Knudsen et al. 2007) and to calculate relevant quantities from the fitted model. For a specific year, we model the expected number of migrants at Hanko Bird Observatory for day i as the sum of two scaled Gaussian distributions

$$E(H_i) = h_1 \text{Normal}(i; m_1, v_1) + h_2 \text{Normal}(i; m_2, v_2) \quad (1)$$

Parameters h_1 and h_2 scale the height of the distribution to the number of migrating birds, m_1 and m_2 are the Gaussian means, while v_1 and v_2 are the corresponding variances. This approach accounts properly for annual variability in the shape of the phenological distribution of migrating woodpeckers, e.g. varying skewness and kurtosis and occasional bimodality. We further assume that the number of observed migrants is negatively binomially distributed (an overdispersed Poisson process) around its expectation, as

$$H_i \sim \text{NegBin}\left(\frac{1}{w}, \frac{1}{1 + wE(H_i)}\right) \quad (2)$$

where w is an overdispersion parameter. For each year this model is fitted separately and in years with less than 20 birds observed, we restrict the parameters to $h_1 = h_2$, $m_1 = m_2$ and $v_1 = v_2$, which is equal to fitting one Gaussian distribution. The models are fitted by searching for the maximum likelihood estimates numerically using the Nelder–Mead simplex algorithm. For all statistical analyses and model fitting purposes we use MATLAB (ver. 7.6.0, R2008a, MathWorks Inc., Natick, MA). Each season specific search is repeated with different starting values 100 times, or until 10 consecutive converged results are produced without improving the previous best fit. All fitted curves are plotted against the data and visually checked to provide reasonably good descriptions of the seasonal variation in migration numbers.

The fitted annual expectation of H is used to calculate the median timing of migration (Med_i) for each year and the expected total number of migrating birds. The latter is used as data describing migration intensity (M_i). Strong correlations between these measures and corresponding measures from raw data indicate that the exact method used is unlikely to affect the results. The Pearson's correlation coefficient between raw (observed) and modelled (expected) annual median migration date is 0.9997 and for annual totals the correlation is 0.9347. For the analysis of migration timing, we exclude years with less than 20 observed woodpeckers (6 of 29 years), because the precision of the estimated timing of peak migration is likely to be weak.

Cone crop data

Data on conifer crops are based on annual cone surveys conducted by the Finnish Forest Research Institute, in the years 1979–2007 (Hokkanen 2000). The cones have been counted in late summer from so-called seed-crop observation stands in southern Finland. The material contains on average 37 spruce stands and 17 pine stands annually. The number of cones of fifty trees with dominant crowns was estimated at each stand. The same tree individuals were considered from year to year, and trees were classified into seven groups according to the abundance of cones (I: no cones, II: 1–20 cones, III: 21–50 cones, IV: 51–100 cones, V: 101–200 cones, VI: 201–500 cones, VII: more than 500 cones). The annual cone abundance indexes are calculated by taking the arithmetic mean of all trees, using the class midpoints (I = 0, II = 10.5, III = 35.5, IV = 75.5, V = 150.5, VI = 350.5, VII = 550.5) for each tree. Even though category VII codes for more than 500 cones, over 600 cones in a tree is rare, whereby the value 550.5 will not introduce any severe bias

into the cone index. Annual pine and spruce cone indices are denoted in the rest of the paper as P_t and S_t , respectively. The indices are presented in Fig. 2. Spruce cone density shows clearly more variability and negative autocorrelation ($\rho = -0.439$, $SE = 0.176$), while that of the pine cones is much more stable and shows a weak positive autocorrelation ($\rho = 0.283$, $SE = 0.188$). On a log-scale the autocorrelations are even more prominent (spruce: $\rho = -0.632$, $SE = 0.152$; pine: $\rho = 0.329$, $SE = 0.185$). The two species' cone indices are in practice mutually uncorrelated ($r = -0.140$, $SE = 0.191$; on log-scale $r = -0.214$, $SE = 0.188$).

The model for migration intensity

We investigate the intensity of irruptive migration with a model that allows for both intrinsic and extrinsic factors acting simultaneously. The expected number of post-breeding (autumn) migrants in year t is modelled as

$$E(M_t) = \alpha N_t^\theta \exp[\beta_1 P'_t + \beta_2 S'_t] \quad (3)$$

N_t is the population density in year t , while P'_t and S'_t are annual indices of pine and spruce cone densities. The parameters of interest are θ that accounts for purely intrinsic effects and the multiplicative effects of cone availability (β_1 and β_2). The parameter α depend on the scales of N , P' and S' , and is hence not of principal biological interest. The model can be thought to consist of two parts: the expected number of migrants based on purely intrinsic factors (αN_t^θ), and an exponential function summarizing the environmental effects ($\exp[\beta_1 P'_t + \beta_2 S'_t]$).

To fit the proposed model statistically to data, the number of migrants (M_t) must be phrased as a stochastic version of the model in Eq. 3. To model the number of migrating individuals, an assumption could be that M_t is Poisson-distributed around its expectation, or negatively binomially distributed (as an overdispersed Poisson-process), or by assuming multiplicative error that is log-normally distributed with an expected value of unity. We use the latter approach that applies also to continuous indices of migration intensity. The model can then be expressed in a useful form on log-scale

$$\ln M_t = \ln \alpha + \theta \ln N_t + \beta_1 P'_t + \beta_2 S'_t + \varepsilon_t - 0.5\sigma^2 \quad (4)$$

where the error term is

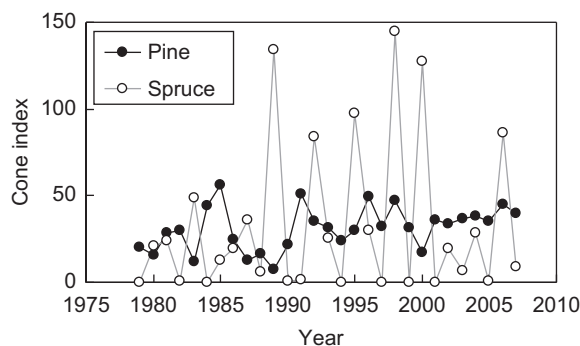


Figure 2. Time series of the pine and spruce cone crop indices. The values correspond approximately to the average number of cones per tree.

$$\varepsilon_t \sim \text{Normal}(0, \sigma^2) \quad (5)$$

In a multiplicative log-normal error distribution with unit mean, the term $-0.5\sigma^2$ is in fact the mean, but by leaving the term outside ε (so that $E(\varepsilon) = 0$) on logarithmic scale, the expression corresponds to an ordinary multiple regression model. Hence, maximum likelihood estimates for the parameters can be obtained using ordinary least squares multiple regression, where the response variable is $\ln M$ and the explanatory variables are $\ln N$ (a proxy of $\ln N$), as well as P' and S' . The fitted intercept will correspond to $\ln \alpha - 0.5\sigma^2$, whereas the regression coefficients are direct estimators of θ , β_1 and β_2 , respectively.

If the expected number of migrants increases proportionally with the population density index, the parameter θ is 1. On the other hand, if density increases the per capita rate of migration, e.g. due to increased competition, the number of potential migrants can be roughly modelled as an exponential function of density ($\theta > 1$). Finally, if the relationship between the breeding population (N_t) and the number of potential migrants is weak, e.g. due to large variation in breeding success, or strong negative density dependence in breeding success, the parameter θ is smaller than one or even close to zero, which means that density does not help in explaining the migration intensity. When fitting the model statistically this situation can also appear if the population size is fairly constant and/or measured with a large amount of observation error. In the set of 12 candidate models (Table 1), we take three different approaches to how density affects the migration intensity: $\theta = 0$, $\theta = 1$ and θ is a free parameter.

The model for population dynamics

Population dynamics in relation to density and resources is evaluated with a set of candidate models, where the expected population density can be described as

$$E(N_{t+1}) = N_t \exp[a_0 + a_1 \ln N_t + b_1 P'_t + b_2 S'_t] \quad (6)$$

Table 1. The 12 competing candidate models for irruptive migration intensity, sorted according to model support (in terms of AIC_c). Models 10–12 are structurally different due to different effects of covariates (P , pine cones; S , spruce cones), while all other models are special cases of these. θ is the effect of population density, and K is the number of estimated parameters. The support for each of the candidate models is presented in terms of the negative log-likelihood ($-\ln L$), differences in AIC_c compared to the best ranked candidate model (ΔAIC_c) and Akaike weights (ω).

Model	Covariates		θ	K	$-\ln L$	ΔAIC_c	ω
(6)	P	S	1	4	41.93	0.00	0.306
(8)	$\ln P$	$\ln S$	1	4	42.09	0.31	0.262
(4)	$\ln P$	$\ln S$	0	4	42.54	1.23	0.165
(2)	P	S	0	4	43.07	2.28	0.098
(12)	$\ln P$	$\ln S$	free	5	41.90	2.88	0.072
(10)	P	S	free	5	41.90	2.89	0.072
(3)	P/Z	S/Z	0	4	45.05	6.23	0.014
(7)	P/Z	S/Z	1	4	45.71	7.55	0.007
(11)	P/Z	S/Z	free	5	45.05	9.17	0.003
(5)	–	–	1	2	51.38	13.71	0.000
(9)	–	–	free	3	51.27	15.97	0.000
(1)	–	–	0	2	52.81	16.55	0.000

The symbols N , P' and S' again stands for population density, pine and spruce cone indices. This is a modified version of the Gompertz-model (Gompertz 1825) of population growth, with additional multiplicative effects of P'_t and S'_t . The parameters of primary interest are the density dependence parameter a_1 , and the multiplicative effects of pine and spruce cone availability b_1 and b_2 . As α in the migration intensity model described in previous section, a_0 is affected by the scales of the involved variables and the effects of P' and S' .

A stochastic version of the population dynamical model is needed for statistical analysis of the data. For this purpose we use a state-space model separating between process and observation error, which is important in order to successfully estimate the strength of density dependence (Freckleton et al. 2006, Knappe 2008). Biases in density dependence would in turn also bias the effects of positively or negatively autocorrelated covariates, such as the density of spruce cones (Lindén and Knappe 2009). We consider the described model on the log-scale. By denoting $\ln N_t$ as X_t the model can be written

$$X_{t+1} = a_0 (1 + a_1)X_t + b_1P'_t + b_2S'_t + \gamma_t \quad (7)$$

$$\gamma_t \sim \text{Normal}(0, \sigma_{\text{proc}}^2) \quad (8)$$

$$\ln Z_t = X_t + \psi_t \quad (9)$$

$$\psi_t \sim \text{Normal}(0, \sigma_{\text{obs}}^2) \quad (10)$$

where γ and ψ are process and observation error, respectively. This corresponds to the linear state-space model, where $\ln Z$ represents the time series data; X is the state variable, whereas P' and S' are covariates in the model.

Here we take three different approaches to density dependence (Table 2). First, we consider candidate models assuming exactly compensatory dynamics, where a_1 is fixed to -1 . In such cases the assumed dynamics are strongly density dependent but the upcoming population density can not be predicted with that in the previous year. It is impossible to separate between process and observation error in these models and hence there is one parameter less to estimate (Knappe 2008). Since the result is invariant on the partitioning of error, we fit these models assuming that there is no observation error. Secondly, we consider candidate models assuming exponential growth, where a_1 is fixed to zero and density as such has no effect on population growth. Since populations can not grow to infinite size, this means in practice that the population is thought to be controlled by other factors to such an extent that intraspecific competition is irrelevant. Finally, we consider candidate models with parameter a_1 free, i.e. the density dependence is estimated.

The state-space models are fitted by maximum likelihood estimation, again using the Nelder–Mead simplex numerical search algorithm (in MATLAB). The starting value of the state variable (X_1) is assumed to have a normal probability distribution with zero mean and unit variance. To get maximum likelihood estimators with good asymptotic behaviour it is necessary to restrict the process and observation error

Table 2. The 15 competing candidate models for population dynamics, sorted according to model support (in terms of AIC_c). Models 12–15 are different due to the effects of covariates (P , pine cones; S , spruce cones; M , migration intensity), while all other models are special cases of these. The parameter a_1 is the density dependence, and K is the number of estimated parameters. The support for each of the candidate models is presented in terms of the negative log-likelihood ($-\ln L$), differences in AIC_c compared to the best candidate model (ΔAIC_c) and Akaike weights (ω).

Models	Covariates		a_1	K	$-\ln L$	ΔAIC_c	ω
(14)	$\ln P$	$\ln S$	free	6	-6.49	0.00	0.947
(9)	$\ln P$	$\ln S$	0	5	-1.91	5.96	0.048
(4)	$\ln P$	$\ln S$	-1	4	2.58	11.99	0.002
(12)	P	S	free	6	0.06	13.11	0.001
(15)	$\ln M$		free	5	2.23	14.23	0.001
(10)	$\ln M$		0	4	4.34	15.52	0.001
(8)	P/Z	S/Z	0	5	2.92	15.61	0.000
(2)	P	S	-1	4	4.64	16.12	0.000
(11)	–		free	4	4.87	16.58	0.000
(6)	–		0	3	6.24	16.61	0.000
(13)	P/Z	S/Z	free	6	2.12	17.22	0.000
(7)	P	S	0	5	4.81	19.39	0.000
(5)	$\ln M$		-1	3	7.64	19.40	0.000
(1)	–		-1	2	9.30	20.23	0.000
(3)	P/Z	S/Z	-1	4	7.11	21.05	0.000

variances to the interior of the parameter space (Dennis et al. 2006). Each search is repeated with different starting values 100 times, or until 10 consecutive properly converged results with such interior solutions is found without improving the previous best fit. The best interior solution was chosen as the parameter estimates, or if no interior solution was found in any of 100 searches, the result with the highest likelihood among all converged searches was chosen as the parameter estimates.

The standard errors of all state-space models were obtained by parametric bootstrap. The dynamics and observation error were simulated using the point estimates of the model parameters and pseudo random numbers, after which the model was fitted to the simulated data. This procedure was repeated 1000 times. The standard deviations of the fitted parameters (over the repetitions) are the bootstrap estimates of standard errors.

Incorporating resource availability

For each of the three approaches to how density affects migration intensity and population dynamics, we evaluate four different hypotheses regarding pine and spruce cone crops and their possible interaction with density. For population dynamics we also propose a fifth hypothesis: that the irruptions per se explain the population dynamics, rather than food shortage. Table 1 and 2 present the resulting sets of 12 and 15 candidate models, for irruptive migration intensity and population dynamics, respectively. Note that most of these models are nested, i.e. special cases of other models (Hilborn and Mangel 1997), and in fact there are only three structurally different models for irruptive migration and four for population dynamics. The differences lie in what covariates are included in the model and how.

The first hypothesis is that there is no effect of cone density (Table 1: model 1, 5 and 9; Table 2: model 1, 6 and 11). In the other approaches the effects of pine and spruce cones

(or migration) are estimated, but they are incorporated in different ways.

The second hypothesis is rate-dependence, which means that the densities of pine and spruce cones as such are the variables of interest, i.e. $P'_t = P_t$ and $S'_t = S_t$ (Table 1: model 2, 6 and 10; Table 2: model 2, 7 and 12). In population dynamics, this corresponds to the classic Lotka–Volterra type of effect (Berryman 2001), and is based on analogy with molecular mass action (Akçakaya et al. 1995). In that case the variables are treated as environmental conditions, or resources that are not really depleted by their consumers.

The third hypothesis is that of ratio-dependence (Akçakaya et al. 1995, Berryman 2001), an idea developed in the context of predator–prey interactions. The response is here thought to be related to the ratio of resources and consumers (Arditi and Ginzburg 1989). These ratios can be thought of as corrected measures of density, i.e. density per resource (Berryman 2001, Lindström et al. 2005), but one can as well talk about the amount of shared resources, i.e. resource items per individual (Arditi and Ginzburg 1989, Akçakaya et al. 1995, McCarthy 1996). We take the latter approach, $P'_t = P_t / N_t$ and $S'_t = S_t / N_t$ (Table 1: model 3, 7 and 11; Table 2: model 3, 8 and 13) as we essentially want to think of measures of resource availability. According to the defenders of ratio-dependent theory, the ratio can be thought to be a better approximation of the realized resource availability on an ecological time-scale (Akçakaya et al. 1995). This approach is worth considering, especially in the cases of renewable resources with highly temporally variable availability (Berryman et al. 1995, Lindström et al. 2005).

The fourth approach is to measure cone availability as logarithm-transformed cone abundance (Table 1: model 4, 8 and 12; Table 2: model 4, 9 and 14), corresponding to an exponential effect on the raw scale. Hence, $P'_t = \ln P_t$ and $S'_t = \ln(S_t + 1)$, the latter due to one year of zero spruce cones counted. Visual investigation of the transformed time-series reveals that the transformation $\ln(S_t + 1)$ is very similar to $\ln S_t$. One aspect of this approach is that the variation is emphasized at smaller cone densities. Increased wintering mortality does not necessarily show a linear relationship to cone availability and the appetite of the woodpeckers is definitely finite. Another important aspect of this approach is that – together with the irruptive migration model with free θ or the Gompertz model with free a_1 – it does not separate between rate and ratio-dependence. For example, the effect of density and spruce on the logarithm of migration intensity can be written as $\theta \ln N_t + \beta_2 \ln S_t$, or alternatively as $(\theta + \beta_2) \ln N_t + \beta_2 \ln(S_t / N_t)$. Although this form of the model is robust for assumptions on rate and ratio dependence, the drawback is that the biologically interesting parameters θ or β_2 are more difficult to interpret.

For the population dynamical models we also investigate the possibility that migration intensity alone act as a better predictor for population dynamics compared to food shortage (Table 2: model 5, 10 and 15). This model also has one parameter less to estimate compared to the corresponding ones with cones as covariates. We here choose to use the log-transformed intensity of irruption, to keep the type of effect similar to the Gompertz density dependence and avoid outlier effects of the strong irruption in 2001. This type of multiplicative effects of irruptions on population dynamical

is plausible if migration affects individual survival, although possible immigration effects will most likely be detected using this parameterization as well.

Statistical inference

We use an information theoretical approach to compare the candidate models of irruptive migration intensity and population dynamics. The performances of the candidate models are evaluated in terms of differences in Akaike information criterion corrected for small sample size (AIC_c) and Akaike weights (ω), which are standard techniques in information theoretical model selection (Burnham and Anderson 2002). The importance of different groups of candidate models is investigated by comparing their summed Akaike weights and comparison of two candidate models is done using evidence ratios (the ratio of two Akaike weights). Parameter uncertainty is considered by relating point estimates to their associated standard errors, acknowledging that the estimates ± 2 SE are approximate 95% CI.

In the approach used here, we assume log-normally distributed and temporally uncorrelated error structures of the models of migration intensity and population dynamics. To check the validity of our assumptions, we test the null hypothesis of normally distributed model residuals (fitted on a log-scale) applying Lilliefors' test (Lilliefors 1967). We also check the residuals for autocorrelation. In both tests we implement the risk level of 5%.

The relationship between median timing of migration (Med) and the migration intensity (M) is described with reduced major axis regression, assuming a log-linear relationship (Med vs $\ln M$). The strength of the relationship is investigated using Pearson's correlation coefficient and a 95% CI for the coefficient is constructed using Fisher's z -transformation.

Results

The highest ranked candidate model for intensity of irruptive migration is model 6, with θ fixed to 1 and cone density included as such (Table 1). This model explains a fairly large part of the variance in $\ln M_t$ ($R^2 = 0.528$) and has an Akaike weight suggesting that the set contains other plausible candidate models ($\omega = 0.306$). The time-series and the fitted values of candidate model 6 are shown in Fig. 3A. The second best candidate model (8), performs in practice equally well ($R^2 = 0.523$, $\omega = 0.262$) and is similar to model 6, but has log-transformed cone density. The candidate models with $\theta = 1$ have a summed Akaike weight of 0.575 (Fig. 4A). We interpret this as an indicative result that woodpecker breeding density is indeed partially determining the intensity of irruptive migration, by increasing the number of potential migrants. There are no clear amplified effects related to density. Due to possible difficulties in interpreting parameter θ in models with logarithm transformed cones we use model 2, 6 and 10 to compare those effects. In candidate model 10, which corresponds to model 6 but has free θ , the estimate of θ is 0.872 (SE = 0.603). The evidence ratio of model 6 against model 10 is 4.24, suggesting some preference for the simpler candidate

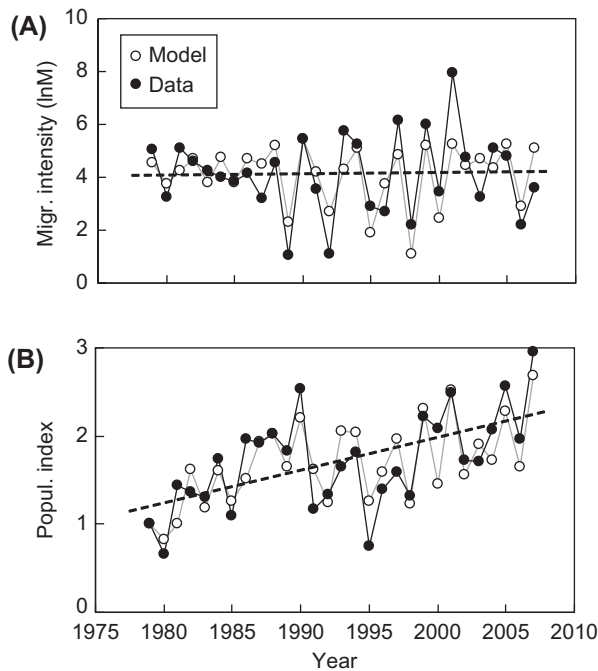


Figure 3. The observed time series (Data) and the predictions according to the highest ranked candidate models (Model) for: (A) the natural logarithm of irruptive migration intensity and (B) population density. In panel (B) the time series of model prediction is actually the estimated state variable (transformed to original scale). Dashed lines indicate least-squares lines fitted to the raw data. The population density index shows an increasing trend of 0.037 (SE = 0.010) per year.

model. The corresponding evidence ratio of model 6 against model 2 without any effect of density ($\theta = 0$) is 3.13, suggesting that including an effect of density is beneficial for explanatory power, but not crucial. The strong effect of lack of cones on migration intensity is evident (Fig. 4B, Table 1), but the functional form of the relationship is not clear. The cone indices as such explain the migration intensity equally well as their natural logarithms. There is no evidence for direct ratio dependent effects. A striking pattern is that the (negative) effect of pine cone crops is weak and does not statistically differ from zero, whereas the negative effect of spruce cone crops is stronger (Table 3). In model 6

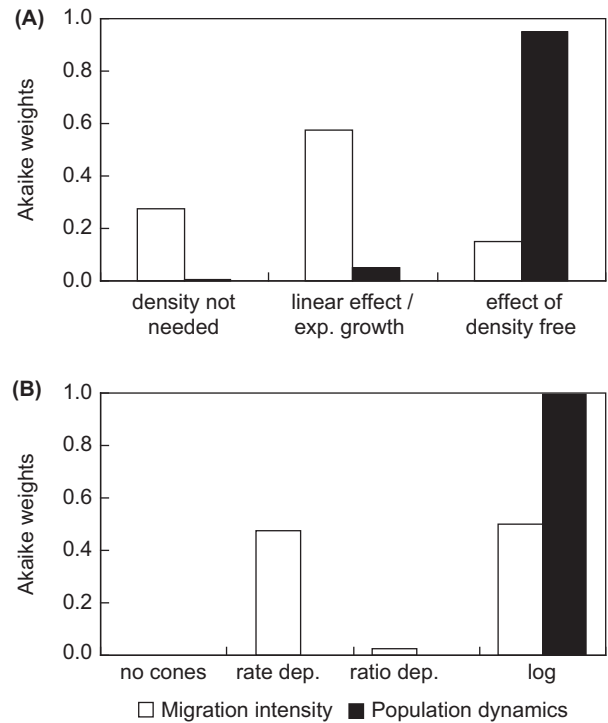


Figure 4. Summed Akaike weights for groups of candidate models with (A) different effects of population density and (B) different effects of cone density (no effect of cones, rate-dependence, ratio-dependence or log-transformed cone density). Candidate models with migration intensity as a predictor instead of cones (population dynamics only) have a negligible summed Akaike weight that is not displayed here. Summed weights for models of irruptive migration intensity are displayed with white, whereas those of population dynamics are with black.

the coefficients are for pine $\beta_1 = -0.015$ (SE = 0.017) and for spruce $\beta_2 = -0.023$ (SE = 0.005). In model 8, with log-transformed cone density the estimated effects are $\beta_1 = -0.114$ (SE = 0.439) and $\beta_2 = -0.583$ (SE = 0.122).

The median timing of migration is negatively related to the logarithm of intensity of migration (reduced major axis regression: $M_t = \exp[18.17 - 0.053 \text{Med}_t]$, SE of slope = 0.01). Thus, a reduction in migration numbers by half (50%) is associated with a 13.1 day delay in median migration date (Fig. 5). The strength of the relationship is

Table 3. Parameter estimates and standard errors (in brackets) for the candidate models of irruptive migration intensity, sorted according to AIC_c (Table 1). Effects of density (θ), cone crops of pine (β_1) and spruce (β_2) that differ from zero by at least two SE in bold. Also the estimated error variances (σ^2) are reported.

Model	Intercept	θ	β_1	β_2	σ^2
(6)	4.857 (0.594)	1	-0.0151 (0.0165)	-0.0228 (0.0047)	1.177 (0.326)
(8)	5.432 (1.574)	1	-0.1144 (0.4394)	-0.5832 (0.1222)	1.190 (0.330)
(4)	6.130 (1.599)	0	-0.1410 (0.4464)	-0.6347 (0.1242)	1.228 (0.341)
(2)	5.424 (0.618)	0	-0.0163 (0.0172)	-0.0241 (0.0048)	1.273 (0.353)
(12)	5.675 (1.651)	0.653 (0.613)	-0.1236 (0.4455)	-0.6011 (0.1278)	1.222 (0.346)
(10)	4.929 (0.695)	0.872 (0.603)	-0.0153 (0.0169)	-0.0229 (0.0048)	1.222 (0.346)
(3)	5.022 (0.467)	0	-0.0130 (0.0205)	-0.0274 (0.0069)	1.459 (0.405)
(7)	4.121 (0.478)	1	0.0041 (0.0209)	-0.0246 (0.0071)	1.527 (0.424)
(11)	5.002 (0.945)	0.023 (0.905)	-0.0126 (0.0260)	-0.0273 (0.0075)	1.518 (0.429)
(5)	3.626 (0.269)	1	—	—	2.098 (0.561)
(9)	3.447 (0.469)	1.372 (0.789)	—	—	2.158 (0.587)
(1)	4.109 (0.282)	0	—	—	2.314 (0.618)

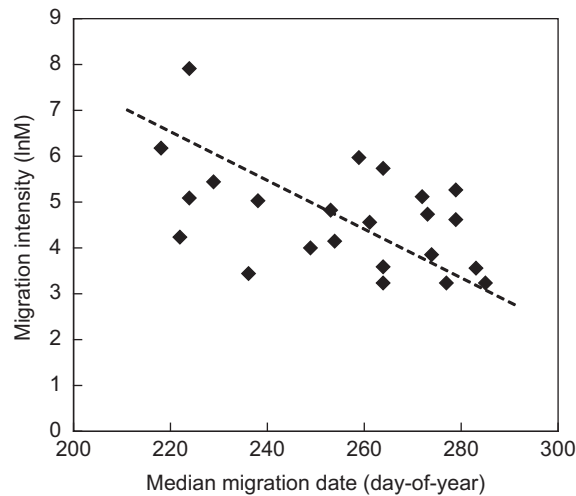


Figure 5. Scatter plot of estimated annual median migration date (Med; date given as day-of-year) and migration intensity (lnM; natural logarithm of number of birds), in years with more than 20 migrating woodpeckers observed. Heavy irruptions tend to occur earlier in autumn compared to weak irruptions. The dashed line is the reduced major axis, illustrating the relationship between the variables.

moderate (Pearson's correlation: $r = -0.479$, $SE = 0.192$, 95% CI = $[-0.744, -0.084]$, $DF = 21$).

The best candidate model of population dynamics is model 14, a Gompertz model with free density dependence and log-transformed cone density and it is superior to the other competing candidate models ($\omega = 0.947$; Table 2). The time-series of the population density index, along with the fitted state variable of model 14 are presented in Fig. 3B. The population index shows an increasing trend, which is satisfactorily captured by the model. There is strong evidence for negative density dependence (Fig. 4A). In model 14 the estimate is $a_1 = -0.348$ ($SE = 0.121$), which corresponds to undercompensatory population dynamics. The positive population dynamical effects of cones are strong. The candidate models including the logarithm of cone indices have a summed Akaike weight of 0.997 (Fig. 4B), suggesting strong

preference for that approach. Like in the case of irruptive migration, spruce cone density is of key importance (Table 2, 4), but the effect sizes are quite similar in pine ($b_1 = 0.131$; $SE = 0.082$) and spruce ($b_2 = 0.158$; $SE = 0.025$). Candidate models that include the logarithm of estimated migration density instead of cones, perform clearly worse than those including cones, although fairly strong negative effects of migration on population growth are evident (Table 4).

We consider the residuals to correspond well to our model assumptions (Appendix 2), particularly so in the inferentially important candidate models. No statistically significant evidence against the null hypotheses of normally distributed and serially independent residuals is found for any of the candidate models of migration intensity. Residual analyses of candidate models of population dynamics are based on the differences between the data $\ln Z$ and the fitted state variable X . For model 15 the null hypotheses of normally distributed residuals can be rejected (Lilliefors' test statistic = 0.190, $p = 0.009$), while model 4 has evidence for first order autocorrelation ($p = 0.489$, $SE = 0.171$, $p = 0.008$). Even if these are the fifth and third best candidate models of population dynamics, they can not be regarded as inferentially important (Table 2, $\omega \leq 0.002$).

Discussion

The intensity of irruptive movements of great spotted woodpeckers is partially determined by population density, such that a more dense population results in a larger number of potential migrants. There is, in other words, no evidence for that high breeding density would increase the per capita migration rate. The most important factor triggering irruptions is failure of spruce cone crops, opposing previous suggestions by Eriksson (1971). Population dynamics show moderately weak negative density dependence and a clear positive effect of spruce cone crops. Pine cone crops do not effectively help in describing the observed patterns of irruption and population dynamics.

The parameter θ that describes the effect of density shows, whenever estimated, a large amount of uncertainty.

Table 4. Parameter estimates and standard errors (in brackets) for the candidate models of population dynamics, sorted according to AIC_c (Table 2). Effects of density (a_1), pine cone crops or irruption intensity (b_1) and spruce cone crops (b_2) that differ from zero by at least two SE in bold. Also the estimated process error variances (σ^2_{proc}) and observation error variances (σ^2_{obs}) are reported.

Model	Intercept	a_1	b_1	b_2	σ^2_{proc}	σ^2_{obs}
(14)	-0.622 (0.290)	-0.348 (0.121)	0.1307 (0.0819)	0.1579 (0.0252)	0.023 (0.013)	0.012 (0.012)
(9)	-0.848 (0.333)	0	0.1421 (0.0894)	0.1687 (0.0348)	0.025 (0.014)	0.016 (0.012)
(4)	-0.097 (0.379)	-1	0.0892 (0.1057)	0.1226 (0.0299)	0.070 (0.017)	—
(12)	-0.051 (0.170)	-0.479 (0.180)	0.0052 (0.0041)	0.0044 (0.0012)	0.041 (0.021)	0.015 (0.018)
(15)	0.640 (0.180)	-0.317 (0.174)	-0.1099 (0.0357)	—	0.038 (0.025)	0.023 (0.021)
(10)	0.527 (0.178)	0	-0.1184 (0.0420)	—	0.035 (0.021)	0.027 (0.019)
(8)	-0.267 (0.106)	0	0.0086 (0.0048)	0.0052 (0.0018)	0.044 (0.021)	0.016 (0.017)
(2)	0.234 (0.155)	-1	0.0042 (0.0044)	0.0040 (0.0012)	0.081 (0.018)	—
(11)	0.113 (0.194)	-0.177 (0.312)	—	—	0.007 (0.021)	0.067 (0.026)
(6)	0.031 (0.018)	0	—	—	0.006 (0.010)	0.068 (0.023)
(13)	-0.074 (0.139)	-0.253 (0.141)	0.0051 (0.0049)	0.0051 (0.0018)	0.051 (0.024)	0.012 (0.020)
(7)	-0.214 (0.141)	0	0.0045 (0.0039)	0.0032 (0.0015)	0.029 (0.019)	0.034 (0.020)
(5)	0.804 (0.171)	-1	-0.0736 (0.0391)	—	0.099 (0.025)	—
(1)	0.500 (0.063)	-1	—	—	0.111 (0.027)	—
(3)	0.534 (0.126)	-1	-0.0060 (0.0056)	0.0037 (0.0018)	0.096 (0.023)	—

This suggests that the identifiability of the parameter is relatively weak and getting more precise estimates might require much more data. Another source of uncertainty is that we here use an estimate of breeding density rather than immediate post-breeding density prior to migration. In our model on irruptive migration it is in fact the latter that is of interest. However, because the relationship between the two densities is likely to be strong, this is probably a minor problem. The lack of spruce cones is the most important factor explaining the intensity of irruptions. The exact functional relationship is not well defined; the spruce cone index as such performs equally well as its logarithm. In contrast to the suggestions of Hildén (1974), cone crops of pine do not predict the large scale irruptive migration of the species.

As has been suggested by Svårdson (1957) and Hildén (1974), more intense irruptions occur earlier in the season compared to minor movements. We suggest that this arises as a result of genuinely earlier onset of migration during intensive irruptions. Annual crops of spruce are usually synchronous over vast ranges (Hokkanen 2001, Broome et al. 2007). Eriksson (1971) showed that annual numbers of migrating great spotted woodpeckers are strongly spatially correlated across Fennoscandia, further reinforcing that the spatial scale of heavy irruptions is large. Like many other irruptive migrants, great spotted woodpeckers have slow migration speed; on average ca 21–25 km day⁻¹ (Hildén and Saurola 1982). Early departure might be necessary in order to first escape the worst areas of crop failure and then establish a wintering territory before winter.

It is known that American and Eurasian tree squirrels use the amount of maturing cones as a cue by which they adjust their reproductive investment to match future food availability (Boutin et al. 2006). Heavy irruptions of great spotted woodpeckers take place already in late summer, a time during which woodpeckers are not dependent on seeds of cones, but feed mainly on insects and plants (Pynnönen 1943, Eriksson 1971). The crucial period for woodpeckers is the winter, during which their diet largely consists of conifer seeds (Cramp 1985). The abundance of fresh spruce cones is evident already in late June, during the breeding season. We believe that the birds predict food shortage much before food is limited, by using the amount of maturing spruce cones as a cue for winter food availability.

The estimated population dynamical model shows weak density dependence, corresponding to slow (undercompensatory) dynamics. The positive trend in density is satisfactorily explained by the best candidate model (Fig. 3B) in that it resembles slow recovery from a population crash. The increase can also be an effect of other factors, such as increased winter survival due to anthropogenic feeding and/or milder winters (Sæther et al. 2000, Francis and Saurola 2004). However, including a trend in the level of population density (carrying capacity) has probably poor statistical identifiability in this short time series, and would hence not introduce much more information. The most likely effect of introducing an additional trend into the model is stronger negative density dependence, i.e. faster fluctuations around the gradually changing carrying capacity.

According to our results, the availability of spruce cones has a strong positive effect on population growth. We find it likely that this effect is due to higher winter mortality and/

or higher rates of natal dispersal (emigration) during extreme lack of spruce cones. This explanation is in concordance with the positive relationship between spruce cone crops and wintering densities of great spotted woodpeckers, which was observed by Eriksson (1971). It has been suggested that the mortality during irruptive movements is in general relatively high (Lack 1954, Bock and Lephtien 1976). Increased mortality is also likely to be related to long migration distances. However, models with migration intensity as a predictor are weaker in performance compared to those with cone crops. Our results might also reflect effects of immigration. A part of the migrating population has inevitably its origin outside our study area. Some of these individuals may settle down in southern Finland during autumns with good cone crops and stay to breed in the following breeding season. Another finding of significance is that population dynamics are best explained by the logarithm of spruce cone crops. This corresponds to a numerical response where the woodpeckers show a finite appetite. Extreme crop failures affect the survival of woodpeckers most strongly, while intermediate crops are not very different from large crops; the woodpeckers do fairly well in both cases. It is also worth noting that the model with the logarithm of the resource is neutral to the question of rate and ratio dependence (Material and methods).

Surprisingly, the availability of pine cones in southern Finland is not an effective predictor in the models, neither for irruptive migration nor for population dynamics, despite the fact that both pine and spruce cones are important winter food resources for great spotted woodpeckers (Cramp 1985). This result is important because it highlights that the ecological role of different resources can differ from the proportion of the resource in the diet of the species. The resource-specific difference is likely to arise as a consequence of the annual variation of the cone crops of the two species: pine has rather stable crops, whereas spruce is a typical masting tree species with high year-to-year variability in crops (Sarvas 1962, 1968, Hokkanen 2000). This view is supported by the fact that the estimated effect sizes of cones on population dynamics are quite similar for pine and spruce. It can be that the great spotted woodpecker uses pine cones and invertebrates lodged under tree bark as a stable food resource, while the extensive variation in spruce cones determines the large scale movements and population dynamics. The result can also be specific for the scale of observation. The spatial distribution of pine cones might affect the wintering distribution of woodpeckers. We suggest further research on the matter.

In spruce there are several insect species that can damage a large part of the seed crops in some years (Annala 1981). In Fennoscandia the most important ones are *Dioryctria abietella*, *Cydia strobilella*, *Hylemyia anthracina*, *Plemeliella abietina* and *Megastigmus strobilobius*. During the worst years of damage the cone density may not represent the amount of seeds available for the woodpeckers very well. Although the density of spruce cones is a good predictor of irruptive movements and population dynamics, the models would probably perform even better, if the effects of pests could be controlled for. Also the used estimates of pine cone crops are likely to be associated with some sampling error, due to worse visibility of pine cones (small size, partly inside the crown), compared to spruce cones.

Irruptive movements have been reported for several animal groups, yet the causal mechanisms for the irruptions are seldom

fully understood. We highlight the need to integrate intrinsic and extrinsic factors in mechanistically sensible models of irruptive migration. Moreover, we emphasize that studying population dynamics and migration behaviour in parallel is often far more informative than separate investigation of those two.

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Appendix 1. The annual time series data used for inferential analyses: population density index (Z), irruption intensity (M), median of migration timing (Med), number of pine- and spruce cones per tree (P and S).

Year	Z	M	Med	P	S
1979	1.000	153.2	238	20.62	0.25
1980	0.659	25.3	264	15.70	20.64
1981	1.433	166.9	272	28.80	23.76
1982	1.371	101.0	279	30.30	0.93
1983	1.302	69.1	222	12.32	48.75
1984	1.745	54.4	249	44.55	0.10
1985	1.098	47.5	274	56.31	13.05
1986	1.971	63.8	254	24.58	19.40
1987	1.923	25.1	285	12.70	35.98
1988	2.022	94.5	261	16.17	5.75
1989	1.827	2.9	209	7.67	134.25
1990	2.542	231.9	229	22.12	1.09
1991	1.171	35.4	283	50.76	1.86
1992	1.329	3.0	273	34.98	84.37
1993	1.644	307.3	264	31.56	25.69
1994	1.808	192.5	279	24.14	0.14
1995	0.749	18.2	258	29.82	97.18
1996	1.390	15.1	256	49.68	30.26
1997	1.591	478.0	218	32.06	0.25
1998	1.315	9.0	238	46.90	145.12
1999	2.223	394.7	259	31.32	0.15
2000	2.080	31.1	236	17.26	127.50
2001	2.489	2766.1	224	35.87	0.00
2002	1.721	114.5	273	33.52	19.38
2003	1.714	25.6	277	37.04	6.69
2004	2.076	163.3	224	38.10	28.33
2005	2.564	124.3	253	35.03	0.50
2006	1.968	9.0	265	45.00	86.60
2007	2.950	36.0	264	40.00	9.30

Appendix 2. Null hypothesis tests for normality (Lilliefors’ test) and serial independence in residuals, for all candidate models of irruptive migration and population dynamics (sorted according to AIC_c). The reported statistics are Lilliefors’ test statistic (L), associated p-values (p_L), first order autocorrelation coefficients (a.c.) and their standard errors in parentheses, as well as associated p-values (p_{a.c.}). Models for which the null hypotheses can be rejected are bolded.

Irruptive migration					Population dynamics				
Model	L	p _L	a.c. (SE)	p _{a.c.}	Model	L	p _L	a.c. (SE)	p _{a.c.}
(6)	0.110	0.484	0.140 (0.194)	0.476	(14)	0.104	0.500	–0.079 (0.196)	0.690
(8)	0.071	> 0.500	–0.163 (0.193)	0.407	(9)	0.144	0.124	–0.081 (0.195)	0.681
(4)	0.103	> 0.500	–0.188 (0.193)	0.339	(4)	0.068	> 0.500	0.489 (0.171)	0.008
(2)	0.089	> 0.500	0.156 (0.194)	0.428	(12)	0.112	0.446	–0.022 (0.196)	0.912
(12)	0.081	> 0.500	–0.186 (0.193)	0.342	(15)	0.190	0.009	–0.080 (0.195)	0.687
(10)	0.113	0.434	0.139 (0.194)	0.481	(10)	0.147	0.104	–0.095 (0.195)	0.632
(3)	0.085	> 0.500	–0.029 (0.196)	0.882	(8)	0.130	0.235	–0.017 (0.196)	0.930
(7)	0.078	> 0.500	0.003 (0.196)	0.988	(2)	0.121	0.331	0.353 (0.184)	0.066
(11)	0.087	> 0.500	–0.029 (0.196)	0.883	(11)	0.142	0.135	–0.054 (0.196)	0.784
(5)	0.096	> 0.500	–0.271 (0.189)	0.162	(6)	0.120	0.337	–0.013 (0.196)	0.948
(9)	0.128	0.246	–0.226 (0.191)	0.248	(13)	0.149	0.094	–0.011 (0.196)	0.955
(1)	0.073	> 0.500	–0.337 (0.185)	0.080	(7)	0.131	0.222	–0.031 (0.196)	0.874
					(5)	0.144	0.123	0.343 (0.184)	0.074
					(1)	0.117	0.378	0.205 (0.192)	0.295
					(3)	0.133	0.201	0.313 (0.186)	0.105